

PBL - PROJECT

FY B.TECH IT SEMESTER II (2024-25)

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SMART

PARKING

A SENSOR BASED APPROACH TO URBAN MOBILITY



NEED STATEMENT

With the increasing number of vehicles, parking space management has become a significant challenge. Traditional parking systems are inefficient, leading to congestion, time wastage, and frustration among drivers. A sensor-based parking system is needed to automate parking management, optimize space utilization, and reduce human intervention.

OBJECTIVES

Automate Parking Space Detection:

- Implement sensors that detect the presence or absence of vehicles in parking slots.
- Ensure real-time monitoring of parking space occupancy.

Provide Real-Time Availability Updates:

- Display available and occupied parking slots on a digital interface or mobile app.
- Enable drivers to check slot availability before entering the parking area.

Reduce Congestion & Optimize Space Utilization:

- Minimize the time drivers spend searching for a vacant spot.
- Improve traffic flow within parking lots.

Ensure a User-Friendly Interface:

- Design a simple and clear display system for easy understanding.
- Potential integration with mobile apps or web-based platforms for remote parking updates.

CONSTRAINTS

Cost-Effectiveness and Scalability:

- The system should be affordable for small and large parking facilities.
- It must be scalable to accommodate different parking lot sizes and structures.

Sensor Range and Accuracy:

- Sensors should be able to accurately detect vehicles within a defined range.
- Avoid false readings caused by environmental factors such as shadows or debris.

Power Consumption and Maintenance:

- Sensors should have low power consumption to ensure long-term operation.
- The system should require minimal maintenance and be durable in different environmental conditions.

FEATURES

Vehicle Detection:

- Sensors will detect whether a parking slot is occupied or vacant.
- Data will be processed and updated in real-time.

Parking Slot Status Display:

- A visual indication (screen display, or mobile app) will show available and occupied spaces.
- Users can access this information before or upon arrival at the parking facility.

Entry and Exit Management:

- The system can potentially track vehicles entering and exiting the parking area.
- It may include an automated barrier control system for restricted parking lots.

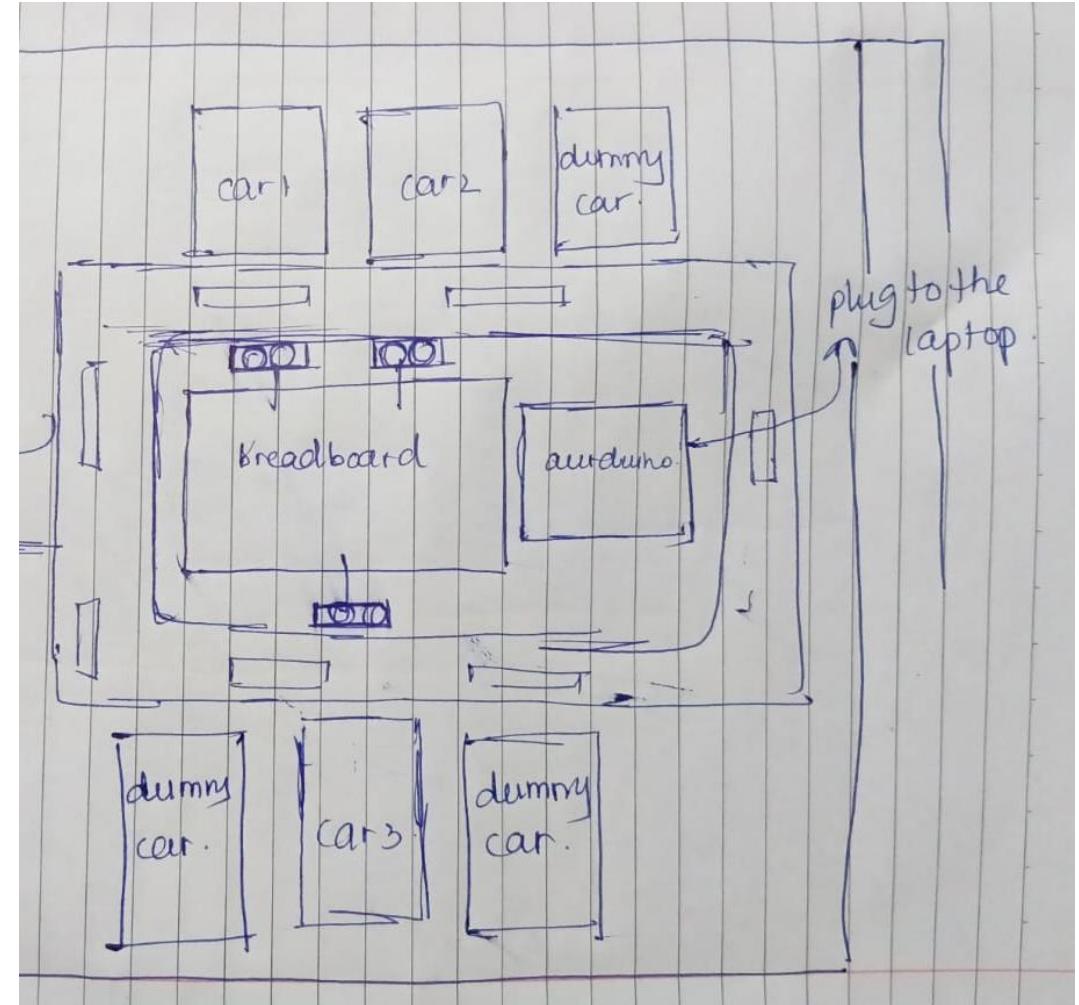
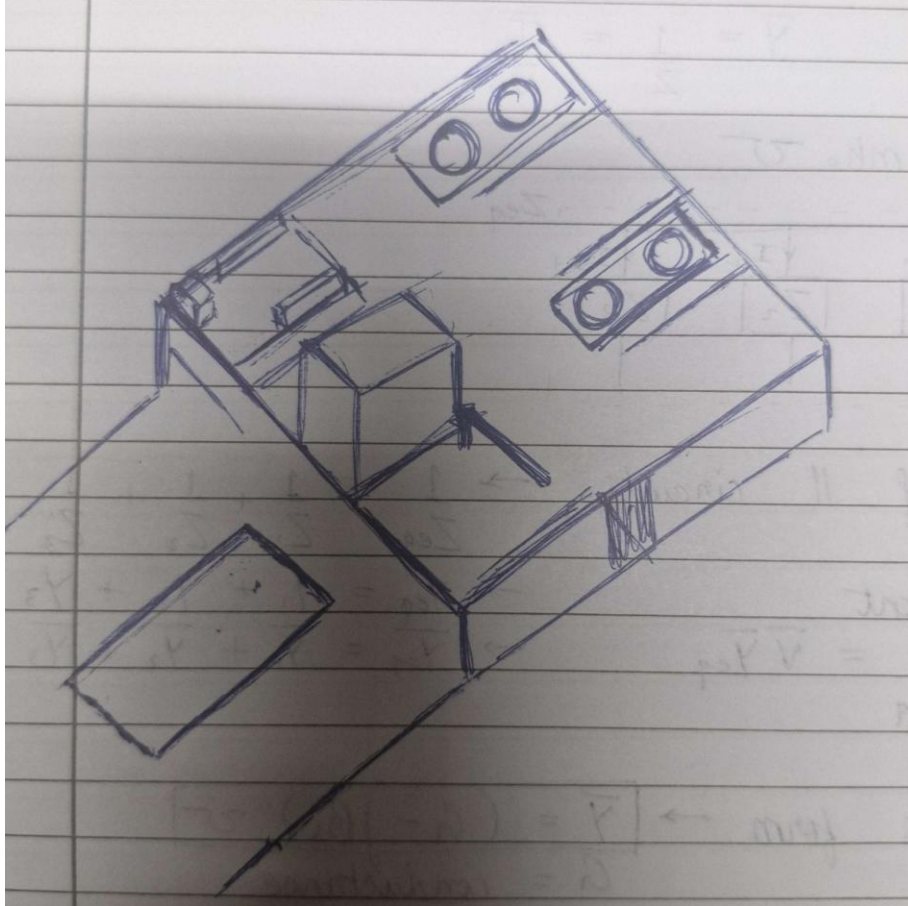
FINAL PROBLEM STATEMENT

The project aims to develop a **sensor-based parking system** that efficiently detects and manages parking space availability. The system should automatically identify vacant and occupied slots using sensors, display real-time information to users, and minimize manual intervention.

SCOPE OF PROJECT

1. **Commercial & Residential Parking** Used in malls, offices, apartments, and gated communities for better space utilization. Enables automated entry, exit, and payment systems.
2. **Traffic Management & Congestion Control** Reduces the time drivers spend searching for parking, easing traffic flow. Helps in designing better road.
3. **IoT & Automation** Uses real-time data from sensors to guide drivers to available spaces.

CONCEPTUAL DESIGN

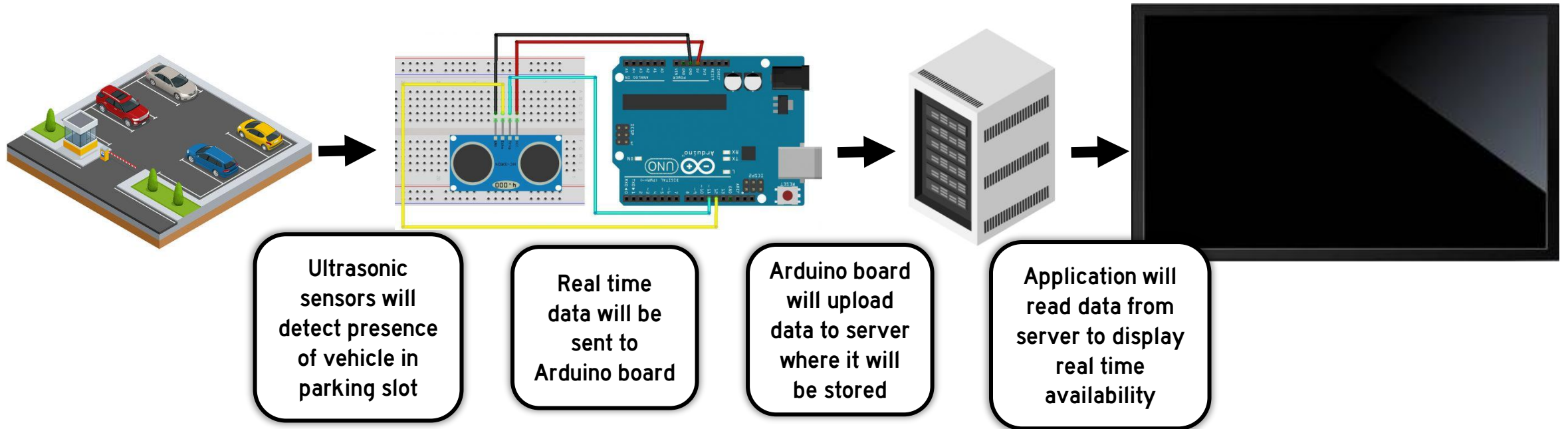


HARDWARE / SOFTWARE REQUIREMENTS

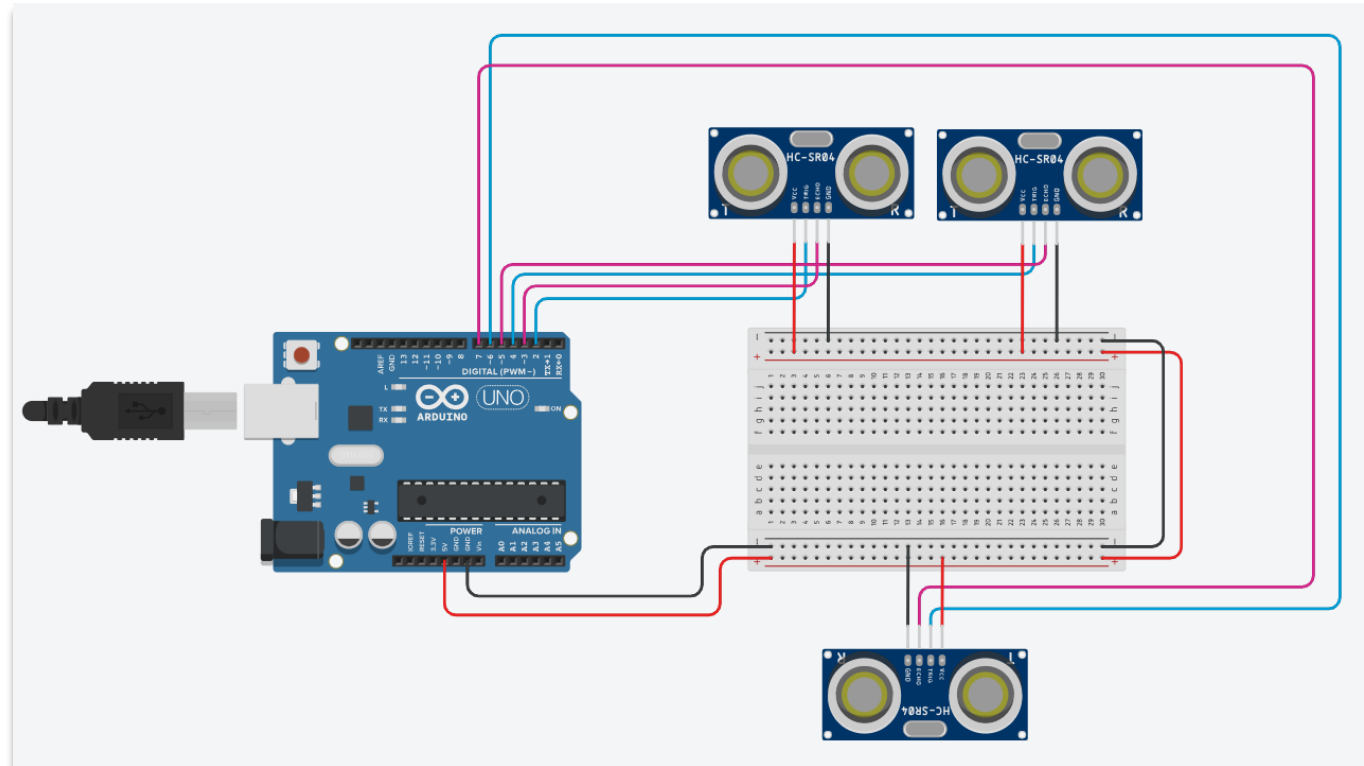
- Arduino Uno R4 — Rs. 990
- Ultrasonic Sensors with mounting brackets — Rs. 59 each (2)
- IR Sensors — Rs. 25 each (3)
- Bread board — Rs. 54
- Jumper wires(M/F & M/M) — Rs. 85
- Cardboard & additional materials
- Arduino IDE
- Approx. area of 300mm x 300mm

UNDER
Rs. 3000!!

SYSTEM FLOW



PROPOSED CIRCUIT/CONNECTION DIAGRAM



BASE CODE

```

1  #define trigPin1 2
2  #define echoPin1 3
3  #define trigPin2 4
4  #define echoPin2 5
5  #define trigPin3 6
6  #define echoPin3 7
7  #define led1 8
8  #define led2 9
9  #define led3 10
10
11 long duration, distance;
12 int lot1_status, lot2_status, lot3_status;
13 const int present = 1;
14 const int absent = 0;
15
16 void setup() {
17     Serial.begin(9600);
18
19     pinMode(trigPin1, OUTPUT);
20     pinMode(echoPin1, INPUT);
21     pinMode(trigPin2, OUTPUT);
22     pinMode(echoPin2, INPUT);
23     pinMode(trigPin3, OUTPUT);
24     pinMode(echoPin3, INPUT);
25
26     pinMode(led1, OUTPUT);
27     pinMode(led2, OUTPUT);
28     pinMode(led3, OUTPUT);
29 }
30

```

```

31 void loop() {
32     lot1_status = checkPresence(trigPin1, echoPin1);
33     lot2_status = checkPresence(trigPin2, echoPin2);
34     lot3_status = checkPresence(trigPin3, echoPin3);
35
36     Serial.print("Lot 1: ");
37     Serial.print((lot1_status == present) ? "Occupied" : "Vacant");
38     led(led1, lot1_status);
39     Serial.print(" | Lot 2: ");
40     Serial.print((lot2_status == present) ? "Occupied" : "Vacant");
41     led(led2, lot2_status);
42     Serial.print(" | Lot 3: ");
43     Serial.println((lot3_status == present) ? "Occupied" : "Vacant");
44     led(led3, lot3_status);
45     delay(100);
46 }
47
48 // Function to trigger sonar sensor and return presence status
49 int checkPresence(int trigPin, int echoPin) {
50     digitalWrite(trigPin, LOW);
51     delayMicroseconds(2);
52     digitalWrite(trigPin, HIGH);
53     delayMicroseconds(10);
54     digitalWrite(trigPin, LOW);
55
56     duration = pulseIn(echoPin, HIGH);
57     distance = (duration / 2) / 29.1;

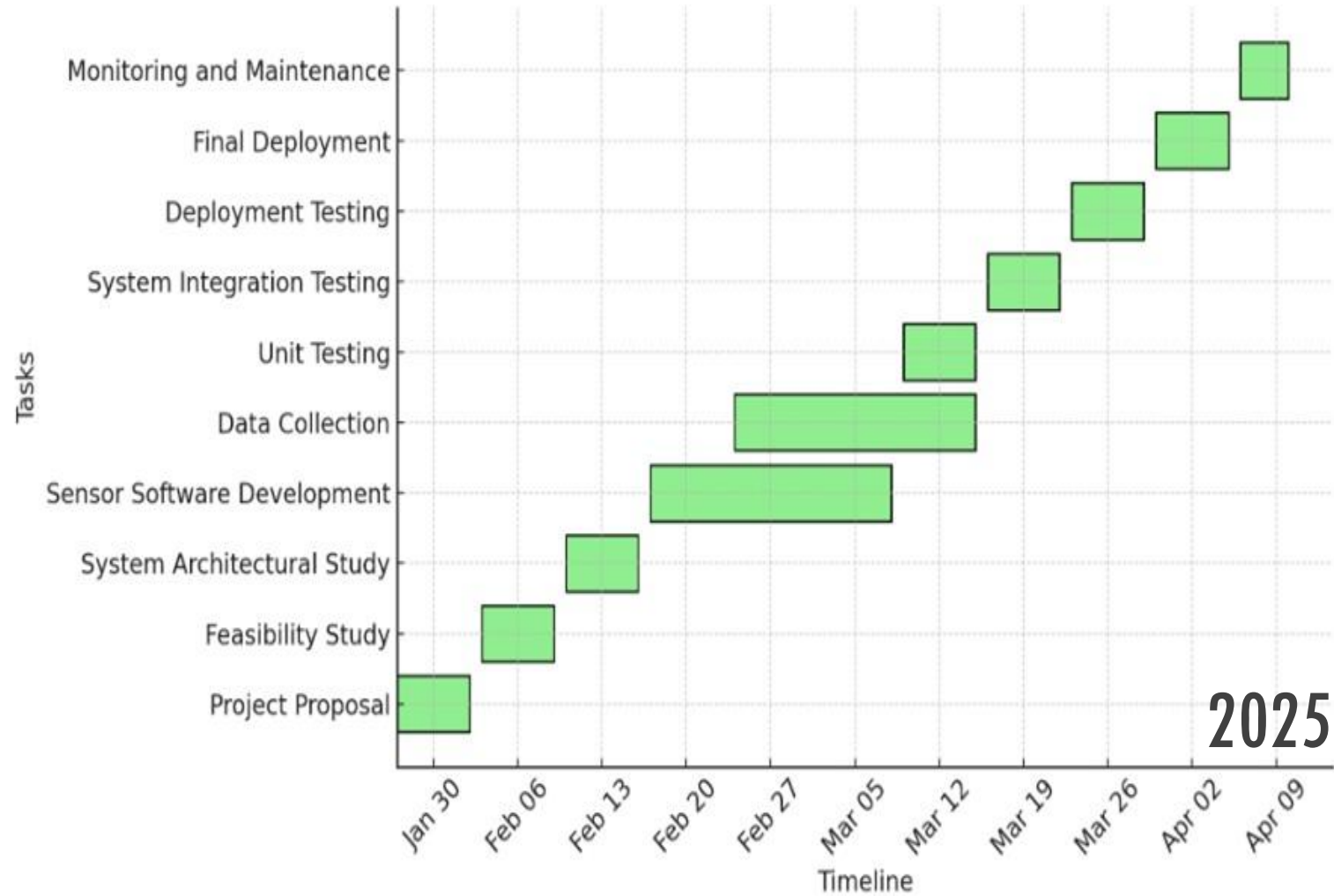
```

```

59     if (distance < 10) {
60         return present;
61     } else {
62         return absent;
63     }
64 }
65
66 void led(int led, int lot_status) {
67     if (lot_status == 1) {
68         digitalWrite(led, LOW);
69     }
70     else {
71         digitalWrite(led, HIGH);
72     }
73 }
74

```


TIMELINE ↑



RESULTS

PROJECT VIDEO



RESULTS

Objective / Function

Achieved / Not Achieved

Remarks

Sensor detection of vacant and occupied spaces

Achieved

85%

Led turning on/off based on the position

Achieved

85%

Locally hosted website

Achieved

100%

REFERENCES →

- <https://howtomechatronics.com/tutorials/arduino/ultrasonic-sensor-hc-sr04/>
- <https://blog.arduino.cc/>
- <https://www.elprocus.com/working-of-different-types-of-motion-sensors/>
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